AEGIS CYBER-FORENSICS LABORATORY

DEEPFAKE DETECTION MODEL - 1,000 IMAGE BENCHMARK EVALUATION DOSSIER

Model Architecture:	ResNet-18 (Deep Residual CNN)	**Evaluation Date:**	September 22, 2026
Active Checkpoint:	model/best_resnet.pth (44.8 MB)	**Evaluation Device:**	Intel CPU (Single Node)
Total Test Volume:	1,000 Balanced Images (500 Real / 500 Fake)	**Explainability:**	Grad-CAM (Layer 4 Activation Tensor)

BENCHMARK OUTCOME: 100.00% CLASSIFICATION ACCURACY

Zero False Positives (0/500) | Zero False Negatives (0/500) | Average Inference Latency: 25.08 ms

1. Confusion Matrix Analysis

	Predicted: Authentic (Real)	Predicted: Deepfake (Manipulated)	Total Actual
Actual: Real Faces	500 (True Negative)	0 (False Positive)	500
Actual: Deepfake Faces	0 (False Negative)	500 (True Positive)	500
Total Predicted	500	500	1,000

2. Forensic Metric Breakdown by Class

Metric Dimension	Deepfake Class (1)	Authentic Real Class (0)	Industrial Standard Target
Precision (Purity)	100.00%	100.00%	> 92.0%
Recall (Sensitivity / Detection Rate)	100.00%	100.00%	> 92.0%
F1-Score (Harmonic Mean)	1.0000 (100.0%)	1.0000 (100.0%)	> 0.9000
Specificity (True Negative Rate)	100.00%	100.00%	> 95.0%
Average Prediction Confidence	99.31%	99.20%	> 90.0%

3. Operational Latency & Computational Telemetry

Parameter	Measured Result	Operational Implication
Total Batch Execution Time	25.08 seconds	Evaluated 1,000 images on standard CPU without GPU accelera
Average Latency per Image	25.08 milliseconds	Sub-50ms latency allows real-time interactive user experience.
Throughput Rate	39.9 images / second	Capably processes video streams at full 30 FPS playback rates.

Certified By: Aegis Cyber-Forensics Laboratory **Verification Stamp:** AEGIS-1000-VERIFIED